Sulabham · APPSC AEE 2026 — Paper 2 (Civil & Mechanical Common)

Module 1: Strength of Materials (from basics)

Notification No. 27/2026 · For Post Codes 1, 2, 3, 4, 6, 7 & 8 · Study weeks 1–3 · Every topic: What it is → Why it happens → Formula → MCQ

How to use this module (3 weeks)

Week 1: Sections 1–3 (forces, stress–strain, elastic constants, strain energy).

Week 2: Sections 4–6 (shear force & bending moment, bending stress, shear stress).

Week 3: Sections 7–10 (principal stress, columns, torsion, springs, cylinders, lifting machines) + all MCQs.

For each section: read the meaning → look at the diagram → copy the formula box into your own notebook → solve the MCQs at the end. Revise the formula boxes every Sunday.

1. Forces, Equilibrium and Friction

What it means: A force is a push or pull. When a body is at rest, all forces on it cancel each other — this is **equilibrium**. Every structure (beam, truss, column) is analysed by first writing the equilibrium equations.

Key formulas

Equilibrium (2-D): **ΣFx = 0, ΣFy = 0, ΣM = 0** (3 equations → 3 unknowns can be found)

Moment = Force × perpendicular distance. Couple = two equal, opposite, parallel forces (produces only rotation).

Lami's theorem (3 concurrent forces in equilibrium): $P/\sin\alpha = Q/\sin\beta = R/\sin\gamma$ (α = angle opposite P).

Truss: members carry only axial force. Solve by **method of joints** (one joint at a time, max 2 unknowns) or **method of sections** (cut ≤ 3 members, take moments).

Perfect truss: **m = 2j − 3** (m = members, j = joints). m < 2j−3 deficient; m > 2j−3 redundant.

Friction: **F = μN**, $\mu = \tan\varphi$ (φ = angle of friction). Angle of repose = angle of friction.

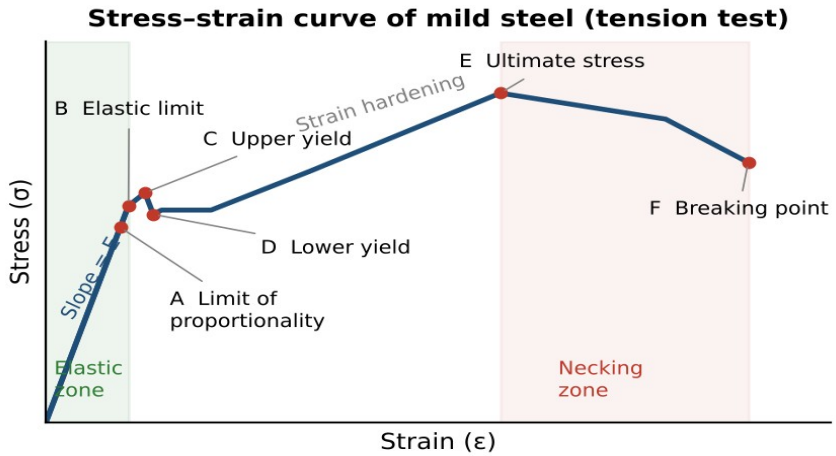
Exam tip

A truss member joining a joint where only two non-collinear members meet with no load is a **zero-force member** — APPSC often asks to spot it.

2. Simple Stresses and Strains

What it means: When you pull a steel rod, its molecules resist. The internal resistance per unit area is **stress (σ)**. The rod also gets longer; change in length ÷ original length is **strain (ε)**. Strain has no unit.

Why it matters: Every design checks that working stress stays below the safe (permissible) stress of the material.



- **Elastic:** load removed → body returns to original shape (zone up to B). **Plastic:** permanent deformation remains.
- **Yield point:** steel suddenly stretches without extra load. Design of steel is based on yield stress (fy).
- **Necking:** after ultimate stress the cross-section reduces locally and the bar breaks. Ductile materials (mild steel) neck; brittle ones (cast iron, concrete) break suddenly without warning.

Key formulas

Stress **σ = P / A** Strain **ε = δL / L** Hooke's law **σ = E·ε** (valid up to limit of proportionality)

Elongation of bar **δL = PL / AE**

Factor of safety = Ultimate (or yield) stress / Working stress (always > 1)

Bar of varying sections: δL = (P/E) Σ (Li / Ai). Tapering circular bar (d1 to d2): **δL = 4PL / (π E d1 d2)**

Bar hanging under self-weight: **δL = ρgL² / 2E = WL / 2AE** (half of same load applied at end)

Composite bar (steel + concrete carry same strain): σs / Es = σc / Ec, P = σsAs + σcAc. **Modular ratio m = Es / Ec**

Temperature stress (ends fixed): **σ = E·α·ΔT** (free expansion δ = αLΔT; no stress if free)

Elastic constants and Poisson's ratio

When a rod is pulled it becomes longer **and thinner**. Lateral strain ÷ longitudinal strain = **Poisson's ratio (μ)**. Three moduli connect stress and strain: **E** (Young's, for tension), **G** (rigidity, for shear), **K** (bulk, for volume change).

Key formulas

E = 2G (1 + μ) **E = 3K (1 − 2μ)** **E = 9KG / (3K + G)**

μ range: **0 to 0.5**. Steel ≈ 0.25–0.3, concrete ≈ 0.15–0.2, rubber ≈ 0.5 (incompressible), cork ≈ 0.

Volumetric strain of a bar (axial load): **εv = ε (1 − 2μ)**. Cube with stress on all three sides: εv = 3σ(1 − 2μ)/E

Number of independent elastic constants: **2** for isotropic material (e.g., E and μ).

Strain energy and types of loading

Strain energy (U) is the work stored inside a body when it is deformed — like a stretched rubber band. **Resilience** = strain energy stored; **proof resilience** = maximum energy up to the elastic limit; **modulus of resilience** = proof resilience per unit volume.

Type of load	Stress produced	Remember
Gradual (slowly applied)	σ = P / A	Normal case
Sudden (applied at once)	**σ = 2P / A**	Twice the gradual stress
Impact (falling from height h)	σ = (P/A)[1 + √(1 + 2AEh / PL)]	If h = 0 → becomes sudden load

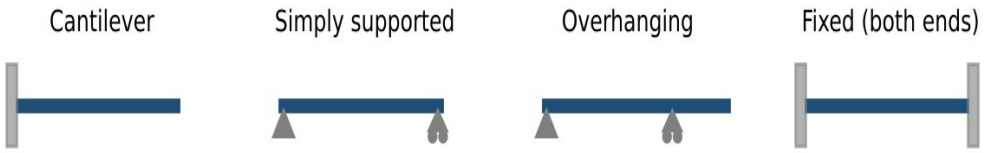
Key formulas

Strain energy **U = σ²·V / 2E** = ½ P·δL Modulus of resilience = σ² / 2E

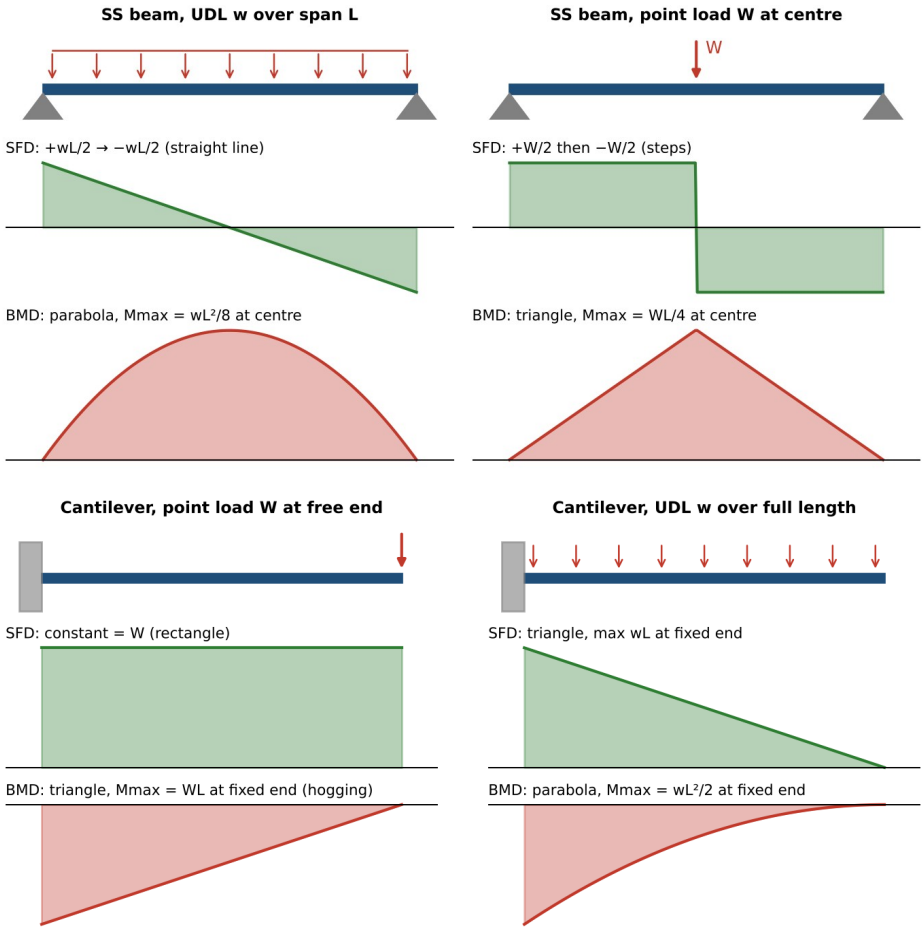
In shear: U = τ²V / 2G

3. Shear Force and Bending Moment

What it means: A beam is a horizontal member that carries loads across a span. At any section, the beam must resist a vertical sliding tendency (**shear force, SF**) and a bending tendency (**bending moment, BM**). Diagrams of these along the length are SFD and BMD.



Sign convention (most books): SF positive if the left part tends to move up. BM positive (**sagging**, beam smiles ∪) ; negative (**hogging**, beam frowns ∩, e.g., cantilever).



Beam and load	Max SF	Max BM	Location
Cantilever, point load W at free end	W	**WL**	Fixed end
Cantilever, UDL w over full length	wL	**$wL^2 / 2$**	Fixed end
Cantilever, UVL (zero at free end to w at fixed)	wL / 2	**$wL^2 / 6$**	Fixed end
SS beam, point load W at centre	W / 2	**WL / 4**	Centre
SS beam, point load W at distance a (b from other end)	Wb/L	**Wab / L**	Under the load
SS beam, UDL w over full span	wL / 2	**$wL^2 / 8$**	Centre
SS beam, UVL zero to w (triangular)	wL/3	**$wL^2 / 9\sqrt{3}$**	0.577L from zero end
Fixed beam, UDL (for comparison)	wL / 2	**$wL^2 / 12$** (ends), $wL^2/24$ centre	Ends

Key formulas

dV/dx = −w (slope of SFD = load intensity) **dM/dx = V** (slope of BMD = shear force)

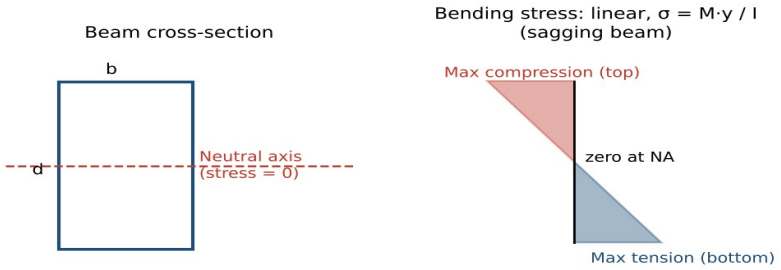
BM is maximum where SF = 0 (or changes sign).

Degree of curve: no load → SF constant, BM linear; UDL → SF linear, BM parabola (2°); UVL → SF parabola, BM cubic (3°).

Point of contraflexure: point where BM changes sign (BM = 0). Occurs in overhanging & fixed beams, never in a simple cantilever or SS beam with downward loads.

4. Bending (Flexural) Stresses

What it means: When a beam sags, the top fibres get shorter (compression) and the bottom fibres get longer (tension). One layer in between neither shortens nor lengthens — the **neutral layer**; its line in the cross-section is the **neutral axis (NA)**, passing through the centroid.



Assumptions of simple bending: material homogeneous & obeys Hooke's law; plane sections remain plane; E same in tension and compression; beam bends in a circular arc with radius large compared to depth.

Key formulas

Bending equation: M / I = σ / y = E / R

Section modulus **Z = I / ymax** → **M = σ·Z** (strength of a section depends on Z)

Rectangle b×d: **$I = bd^3/12$, $Z = bd^2/6$** Circle d: **$I = \pi d^4/64$, $Z = \pi d^3/32$**

Hollow circle: $I = \pi(D^4 - d^4)/64$ Triangle (about centroid): $I = bh^3/36$

Parallel axis theorem: **$I = I_G + A \cdot h^2$**

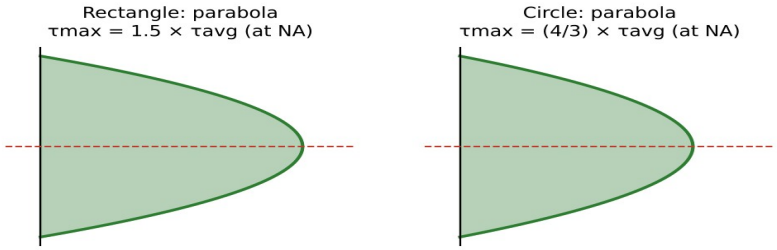
Exam tip

For the same area, an **I-section** is the most efficient beam because material is far from the NA (large Z).

Strongest rectangular beam cut from a round log of diameter D: **d/b = √2**. Stiffest: d/b = √3.

5. Shear Stresses in Beams

Shear force also produces shear stress across the depth. Unlike bending stress, shear stress is **zero at top and bottom** and **maximum at the neutral axis** (exception: some special sections).



Key formulas

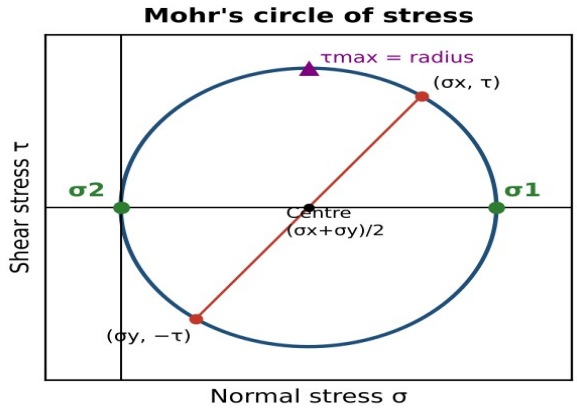
τ = V·A·ȳ / (I·b) (A·ȳ = first moment of area above the level; b = width at that level)

τavg = V / A. Rectangle: **τmax = 1.5 τavg**. Circle: **τmax = 4/3 τavg**. Triangle: τmax = 1.5 τavg at h/2 (not at NA).

I-section: most shear (≈ 95%) is carried by the **web**; most bending moment by the **flanges**.

6. Principal Stresses, Mohr's Circle and Theories of Failure

What it means: At a point inside a loaded body, stresses depend on the plane you look at. Some planes carry **only normal stress with zero shear** — these are **principal planes**, and the stresses on them are **principal stresses (σ1 max, σ2 min)**. Principal planes are 90° apart; planes of max shear are at **45°** to them.



Key formulas

$\sigma 1, \sigma 2 = (\sigma x + \sigma y)/2 \pm \sqrt{[\,((\sigma x - \sigma y)/2)^2 + \tau^2\,]}$

$\tau max = \sqrt{[\,((\sigma x - \sigma y)/2)^2 + \tau^2\,]} = (\sigma 1 - \sigma 2)/2$ = radius of Mohr's circle

Principal plane angle: **$\tan 2\theta = 2\tau / (\sigma x - \sigma y)$**

Uniaxial σ: max shear = **σ/2** on 45° plane. Pure shear τ: principal stresses = **±τ** at 45°.

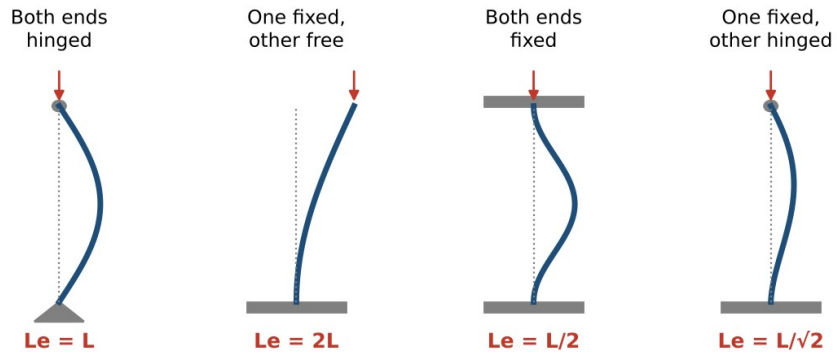
Mohr's circle: centre at ((σx+σy)/2, 0); for like stresses with no shear, circle radius = (σx − σy)/2.

Theory	Also called	Best suited for
Maximum principal stress	Rankine	**Brittle** materials (cast iron)
Maximum shear stress	Guest / Tresca	**Ductile** materials (conservative), design of shafts
Maximum principal strain	St. Venant	—
Total strain energy	Haigh	—
Maximum shear strain energy (distortion energy)	Von Mises / Hencky	**Ductile** — most accurate

7. Columns and Struts

What it means: A vertical compression member is a **column**; an inclined one (in trusses) is a **strut**. Short columns fail by **crushing**. Long, slender columns fail by **buckling** (bending sideways) at a load much smaller than the crushing load. Euler's theory gives this buckling load.

Effective length (Le) of columns — Euler: $P = \pi^2 EI / Le^2$



Key formulas

Euler's crippling load $Pe = \pi^2 EI / Le^2$ — use the **least I** (column buckles about the weaker axis)

Radius of gyration **$k = \sqrt{(I/A)}$** Slenderness ratio **$\lambda = Le / k$**

Euler is valid only for **long columns** (λ large; for mild steel λ > about 80). It ignores direct crushing.

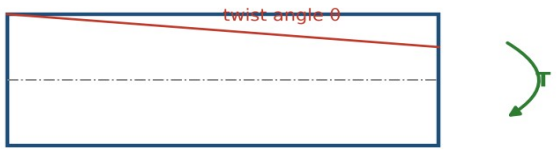
Rankine's formula: $1/P = 1/Pc + 1/Pe$ → $P = fc \cdot A / [1 + a(Le/k)^2]$ (for all lengths; a = Rankine's constant)

Ratio of buckling loads (same L, E, I) — hinged:fixed-free:fixed:fixed-hinged = **1 : 1/4 : 4 : 2**

8. Torsion of Circular Shafts

What it means: A shaft transmits power by twisting (e.g., a motor driving a pump). The twisting moment is **torque (T)**. It produces shear stress which is zero at the centre and maximum at the outer surface.

Torsion: $T/J = \tau/R = G\theta/L$ (τ = 0 at centre, max at surface)



Key formulas

Torsion equation: $T / J = \tau / R = G\cdot\theta / L$

Polar moment of inertia: solid **$J = \pi d^4/32$**; hollow **$J = \pi(D^4 - d^4)/32$**

Torque from max shear: solid **$T = (\pi/16)\cdot\tau\cdot d^3$**

Power $P = 2\pi NT / 60$ (watts; N in rpm, T in N·m)

Torsional stiffness = $T/\theta = GJ/L$. Polar modulus $Z_p = J / R$.

Combined bending + torsion: **$T_e = \sqrt{(M^2 + T^2)}$** (equivalent torque), **$M_e = \frac{1}{2}[M + \sqrt{(M^2 + T^2)}]$** (equivalent moment)

Exam tip

A **hollow shaft** is stronger and stiffer than a solid shaft of the same weight — material near the centre carries little stress.

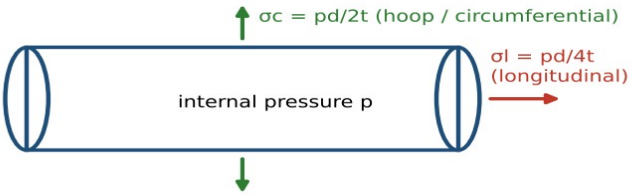
Springs

- **Closely coiled helical spring** (axial load W, mean radius R, wire dia d, n turns): wire is in torsion. Deflection **$\delta = 64WR^3n / (Gd^4)$**, stiffness $k = W/\delta$.
- Springs in **series**: $1/k = 1/k1 + 1/k2$ (softer). In **parallel**: $k = k1 + k2$ (stiffer).
- **Leaf (laminated) springs** are used in vehicle suspensions; they carry bending.

9. Thin Cylinders and Spherical Shells

What it means: Water pipes, boilers and gas tanks carry internal pressure p. If the wall thickness t is small (**t < d/20**), stress is taken as uniform across the wall — a **thin** shell. The pressure tries to burst the cylinder along its length (hoop stress) and pull its ends apart (longitudinal stress).

Thin cylinder: hoop stress = 2 × longitudinal stress



Key formulas

Hoop (circumferential) **$\sigma c = pd / 2t$** Longitudinal **$\sigma l = pd / 4t$** → **$\sigma c = 2\,\sigma l$** (pipes burst along their length)

Max in-plane shear = $(\sigma c - \sigma l)/2$ = **$pd / 8t$**

Volumetric strain of cylinder = **$(pd / 4tE)(5 - 4\mu)$** Sphere: stress = **$pd / 4t$** (all directions), $\varepsilon v = (3pd / 4tE)(1 - \mu)$

Thick cylinders use **Lamé's equations** (stress varies across the wall; max hoop stress at inner surface).

10. Lifting Machines and Virtual Work

A lifting machine (screw jack, pulley block, winch) lets a small effort P lift a large load W.

Key formulas

Mechanical advantage MA = W / P **Velocity ratio VR = distance moved by effort / distance moved by load**

Efficiency η = MA / VR Ideal machine: $\eta = 100\%$, MA = VR

Law of machine: P = mW + C (m = slope, C = effort lost in friction)

Self-locking (irreversible) machine: $\eta < 50\%$ — the load does not come down when effort is removed (e.g., screw jack).

Maximum MA = 1/m; maximum efficiency = 1 / (m·VR)

Principle of virtual work: for a body in equilibrium, total work done by all forces for a small virtual displacement = 0.

Practice MCQs (APPSC AEE pattern)

Attempt without looking at the key. Wrong answers lose 1/3 mark — mark a question only if you can eliminate at least two options.

1. The ratio of lateral strain to longitudinal strain is called
(a) Modulus of elasticity (b) Poisson's ratio (c) Bulk modulus (d) Modular ratio

2. The relation between E, G and μ is
(a) $E = 3G(1-2\mu)$ (b) $E = 2G(1+\mu)$ (c) $E = 2G(1-\mu)$ (d) $E = 9G/(3+G)$

3. Stress produced by a suddenly applied load compared to the same load applied gradually is
(a) Same (b) Half (c) Twice (d) Four times

4. A steel rod with both ends fixed is heated by ΔT. The stress developed is
(a) αΔT (b) EαΔT (c) EαΔT/2 (d) Zero

5. Maximum BM of a simply supported beam with UDL w over span L is
(a) $wL^2/2$ (b) $wL^2/4$ (c) $wL^2/8$ (d) $wL^2/12$

6. Maximum BM of a cantilever with point load W at free end is
(a) WL/4 (b) WL/2 (c) WL (d) 2WL

7. Bending moment is maximum at a section where shear force is
(a) Maximum (b) Minimum (c) Zero or changes sign (d) Equal to load

8. Point of contraflexure is a point where
(a) SF changes sign (b) BM changes sign (c) SF is maximum (d) BM is maximum

9. Section modulus of a rectangular section b × d is
(a) $bd^3/12$ (b) $bd^2/6$ (c) $bd^2/12$ (d) $bd^3/6$

10. For a rectangular beam, maximum shear stress is ___ times the average shear stress
(a) 1.0 (b) 1.33 (c) 1.5 (d) 2.0

11. In a member under uniaxial stress σ, the maximum shear stress is
(a) σ (b) σ/2 (c) 2σ (d) σ/4

12. The theory of failure most suitable for brittle materials is
(a) Max shear stress (b) Max principal stress (c) Distortion energy (d) Max strain energy

13. Effective length of a column fixed at one end and free at the other is
(a) L (b) L/2 (c) 2L (d) L/√2

14. Euler's buckling load of a column fixed at both ends is ___ times that of a column hinged at both ends
(a) 2 (b) 4 (c) 1/4 (d) 16

15. Polar moment of inertia of a solid shaft of diameter d is
(a) $\pi d^4/64$ (b) $\pi d^4/32$ (c) $\pi d^3/16$ (d) $\pi d^3/32$

16. Power transmitted by a shaft rotating at N rpm with torque T is
(a) 2πNT (b) 2πNT/60 (c) πNT/60 (d) NT/60

17. In a thin cylinder under internal pressure, hoop stress is
(a) Equal to longitudinal stress (b) Half (c) Twice (d) Four times the longitudinal stress

18. A lifting machine is self-locking when its efficiency is
(a) More than 50% (b) Less than 50% (c) Exactly 100% (d) More than 75%

19. Number of members in a perfect truss with j joints is
(a) 2j + 3 (b) 2j − 3 (c) 3j − 2 (d) j − 3

20. The volumetric strain of a bar under axial tensile strain ε is
(a) ε(1 + 2μ) (b) ε(1 − 2μ) (c) 3ε (d) ε(1 − μ)

Answer key with one-line reason

1 (b) definition. 2 (b) $E = 2G(1+\mu)$. 3 (c) $\sigma = 2P/A$. 4 (b) $\sigma = E\alpha\Delta T$. 5 (c) $wL^2/8$.
6 (c) WL at fixed end. 7 (c) $dM/dx = V = 0$. 8 (b) BM changes sign. 9 (b) $Z = I/y = bd^2/6$. 10 (c) 1.5.
11 (b) on 45° plane. 12 (b) Rankine. 13 (c) 2L. 14 (b) $Le = L/2 \rightarrow$ load × 4. 15 (b) $\pi d^4/32$.
16 (b) 2πNT/60. 17 (c) pd/2t vs pd/4t. 18 (b) < 50%. 19 (b) $m = 2j - 3$. 20 (b) ε(1 − 2μ).

After finishing this module

1. Score 15+/20 above? Move to Module 2 (Fluid Mechanics). Below 15? Re-read the sections of the wrong answers.
2. Solve 100 more SOM MCQs from APPSC AEE previous papers and ESE/GATE civil question banks.
3. Copy all formula boxes onto 2 sheets — this is your revision sheet for the last week before the exam.